CS 2160: Computer Organization & Assembly Language Programming

Fall 2026

Section: CS 2160-**004**	**Section:** CS 2160-**005**
Time: (**MoWe**) 4:45PM - 6:00PM	**Time:** (**TuTh**) 4:45PM - 6:00PM
Classroom: Kraemer Family Library (**KFL 239**)	**Classroom:** Centennial Hall (**CENT 191**)
Instructor: Himon Thakur	**TA:** Hassan Shakil
Email: hthakur@uccs.edu	**Email:** hshakil@uccs.edu
Office: OCSE 340	**Place:** Remote
Hours: Monday 3:00 PM - 4:00 PM	**TA Hours:** By appointment

Course Description:

Learn the core principles of computer architecture, digital logic, and arithmetic. This course covers how information moves and executes within a system, transitioning into low-level assembly programming. Key topics include addressing modes, array manipulation, control structures, subroutines, parameter passing, and integrating assembly code with high-level languages.

Prerequisite:

CS 1450 or *GDD 2200* (Data Structures and Algorithms) with a grade of **C** or better and *CS 2060* (Programming with C) with a grade of **C** or better.

Understand basic data structures (e.g., arrays, linked lists). Design and implement modular programs in a high-level language (**e.g.**, C++, Java, Python). Write programs in C using functions, arrays, strings, and pointers.

Materials:

Computer Organization and Design RISC-V Edition. *2nd Edition*; David Patterson & John Hennessy; Elsevier. 2021; ISBN 9780128203316; E-book ISBN: 9780128245583;

https://github.com/vanisitor/Computer-Science-Textbooks/blob/master/Computer-Organization-and-Design-The-Hardware-Software-Interface-RISC-V-Edition.pdf. Previous editions of the books are obsolete.

You can access this resource through the VitalSource tab in the course **Canvas** page.

Objectives

Through this course you will understand the importance of mathematical systems in hardware design, how high level programming languages relate to computer hardware, and how data are represented within a computer system. You will also obtain skills in assembly language programming and its influence on high-level languages

and data structures. Most important, you will be able to reason about software design in relation to computer hardware.
The skills and knowledge obtained in this course will enable you to:

1. Understand the basic composition of computer systems
2. Understand Boolean algebra
3. Understand digital logic
4. Understand the relationship between mathematics and computers
    - Representations of data types and formats in computer systems
    - Implementation of arithmetic and logic operations in computer systems
5. Interact with computers at the machine level
    - Understand the composition of assembly instructions
    - Understand the execution of instructions by processors
    - Understand the basic structure of assembly programs
    - Construct programs using assembly language

Topics

1. Introduction to computer systems
    - History and trends
    - The composition of computer systems
    - Commonly used terminology
2. Data representation in computers
    - Signed and unsigned integers
    - Floating point numbers and their representations
3. Computer arithmetic
    - Operations of addition, subtraction, multiplication, and division
4. Boolean algebra and digital logic
    - Basics of Boolean algebra
    - Combinational logic
    - Sequential logic
5. Assembly programming
    - Operations of computer hardware
    - Instruction formats
    - Categories of instructions
    - Hardware support for call-and-return procedures
    - Addressing modes

All course topics will come from Chapters 1 - 3 and Appendices A and C.

Grading:

Grades will be computed by the following weights.

Grade	Percentage
A	$[90, 100]$
B	$[80, 89]$
C	$[70, 79]$
D	$[60, 69]$
F	$[0, 60]$

- Homework assignments $\times$ 4 = **25%**
- Midterm Exam = **25%**
- Coding project assignments $\times$ 4 = **25%**
- Final Exam = **25%**

Final curving will be applied to the final semester grade.

Deliverables:

Submission guidelines:

All assignments will be electronically submitted through the Canvas portal by 11:59 PM on the posted due date. In your submission, please make it clear what your answer to the problem is (e.g. a calculation or the output of a program), and show the necessary logic, work, calculations, or code that you did to arrive at your answer. For problems (especially projects) involving creating code, please make the code easily accessible so that the grader may simply copy and paste to run it themselves.
Late submissions will not be accepted.

Homework

There will be homework problem sets assigned every 1-2 weeks. They will often be a mixture of theory and coding problems. You must complete the homework assignments on your own. This means your submitted answers to the homework assignments are completely written on your own (i.e. calculations and code).

When you put your name on your work, you are the author of that work. Copying code is plagiarism if the original source is not cited, and copyright infringement if permission was not granted to copy the code. Even if permission was granted (e.g. open-source licenses) and the original work is cited, the policy in this class prohibits using code you did not author, and hence no credit will be awarded for that portion of the assignment.

The homework and project assignments are designed to take a lot of time to complete. Most of the learning will happen as you complete the assignments (and the associated readings) *on your own*, and not during the lectures themselves.

Coding project assignments:

Coding projects will focus on developing working programs using the RISC-V assembly language. We will use a free web browser-based RISC-V environment found here: https://comparch.edu.cvut.cz/qtrvsim/app/. We will use this tool throughout the semester. There will be four coding projects, each one taking 2-3 weeks to complete.

Exams

We will plan to have a midterm and a final exam. They will take place in the classroom, and will cover materials from the lectures, homeworks, project assignments, and textbook. The exams will be closed-book and closed-notes, but students will be permitted to bring one sheet of double-sided handwritten or printed notes. Make-up

exams will not be permitted, unless permission is obtained from the instructor at least 24 hours before the exam date. There will be no re-takes for full exams; the grade you earned is the grade that will be recorded.

Participation & Bonus Points

This part of your grade will be based on your overall class attendance, and the timeliness of your homework and project submissions throughout the semester. Bonus points might be added to your final grade based on your performance (e.g., submitting deliverables using LaTeX, etc.).

Policies on the use of AI tools

The use of any AI tool for generating final answers for any deliverables is prohibited.

Cheating and Plagiarism

Absolutely no cheating, copying, or plagiarizing on homework assignments, exams, and summaries. Cheating will result in a grade of an automatic zero for the entire deliverable, and repeated cheating will result in a failing grade for the course. For further details on academic honesty the student is referred to the University Catalog.

Conduct

Students are expected to abide by the UCCS Student Code of Conduct
`http://www.uccs.edu/dos/student-conduct.html`

Accommodations for Disabilities

If you are a student with a disability and believe you will need accommodations for this class, it is your responsibility to register with Disability Services and provide them with documentation of your disability. They will work with you to determine what accommodations are appropriate for your situation. For any accommodations, submit the accommodation request to the instructor at least one week in advance. The accommodations are not retroactive and the disability accommodations cannot be provided until a Faculty Accommodation Letter has been given to the instructor. Please contact Disability Services for more information at Main Hall 105 (dservice@uccs.edu, 719-255-3354)

Military Students

If you are a military student with the potential of being called to military service or training during the course of the semester, you are encouraged to contact your UCCS course instructor no later than the first week of class to discuss the class attendance policy. Please see the following for more information: `http://www.uccs.edu/ military`